Manufacturing Downtime Analysis

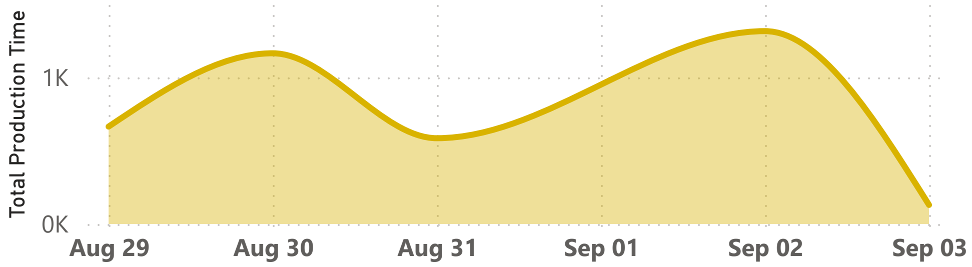
Production Performance Overview:

This provides an analysis of total operating time, recorded downtime, and a comparison between achieved performance and expected benchmarks.

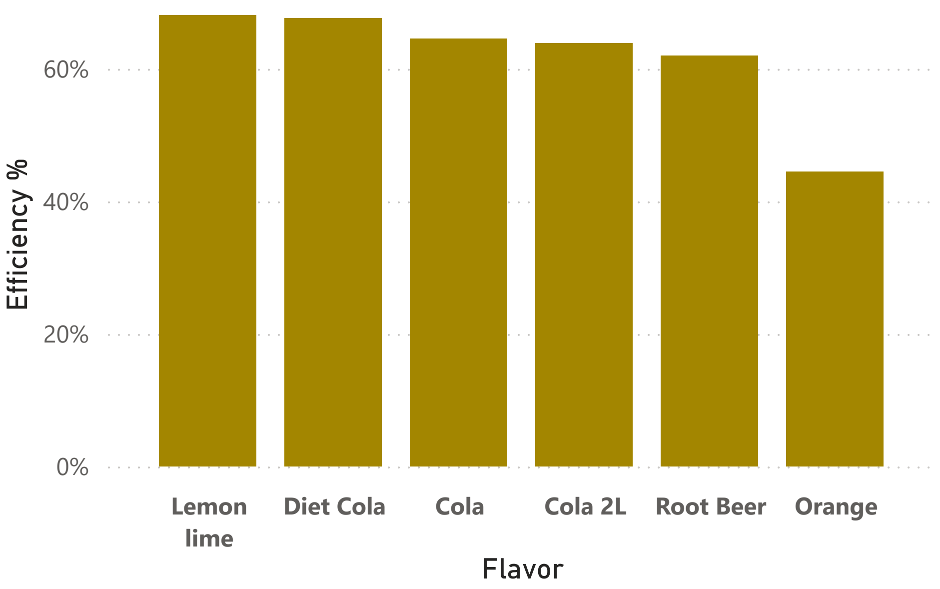
Period Covered: 29 August - 3 September 2024

Total Downtime (Mins)	Total Production time (Mins)	Net Operating Time (Mins)	Efficiency (%)
1388	3858	2470	64%

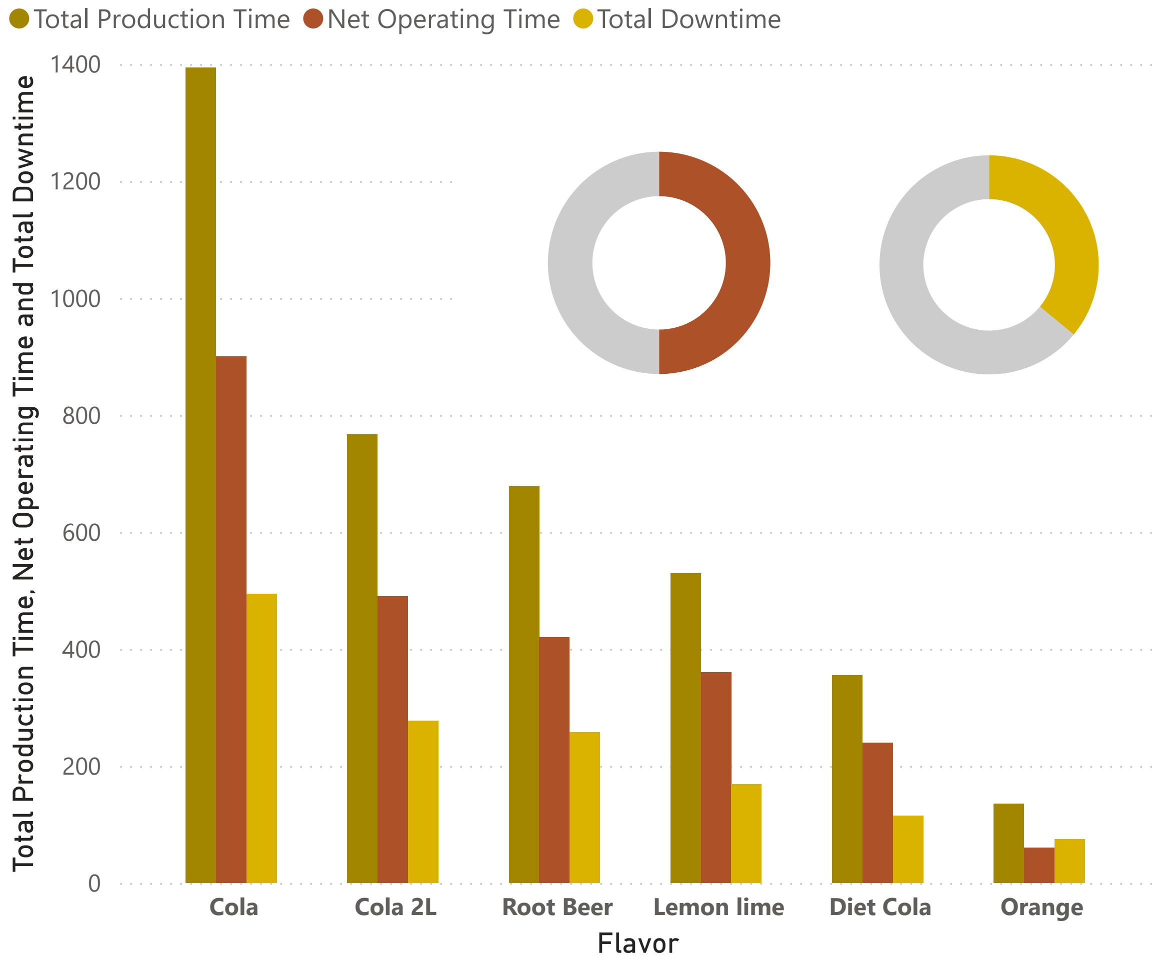
Production Time by Date



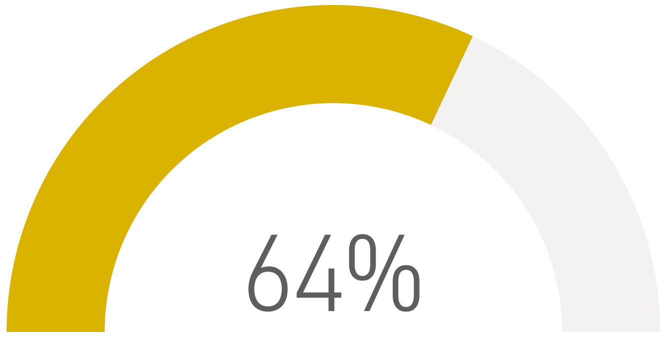
Efficiency % by Flavor



Production Time, Net Operating Time and Total Downtime by Flavor



Line Efficiency



From **29 August to 3 September**, total production time was **3,858 mins (64.13 hrs**), with **2,470 mins (41.17 hrs)** as net operating time and **1,388 mins (23.13 hrs)** lost to downtime, resulting in an **efficiency of 64%**.

Production fluctuated over the period. **Lemon Line and Diet Cola recorded the highest efficiency (68%)**, followed by **Cola (65%) with Orange recording the lowest efficiency of 44%**. **Cola also had the highest production time and downtime**, followed by **Cola 2L**.

Overall, the analysis shows that **higher production time is associated with increased downtime**, indicating that high-volume lines may require better maintenance and optimization to improve efficiency.

Downtime Root Cause Analysis

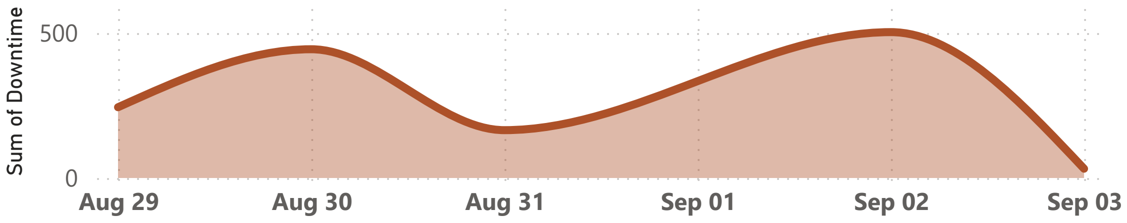
Production Constraints Analysis:

This evaluates the sources of downtime, their frequency, and determines whether disruptions are due to operator actions or process inefficiencies.

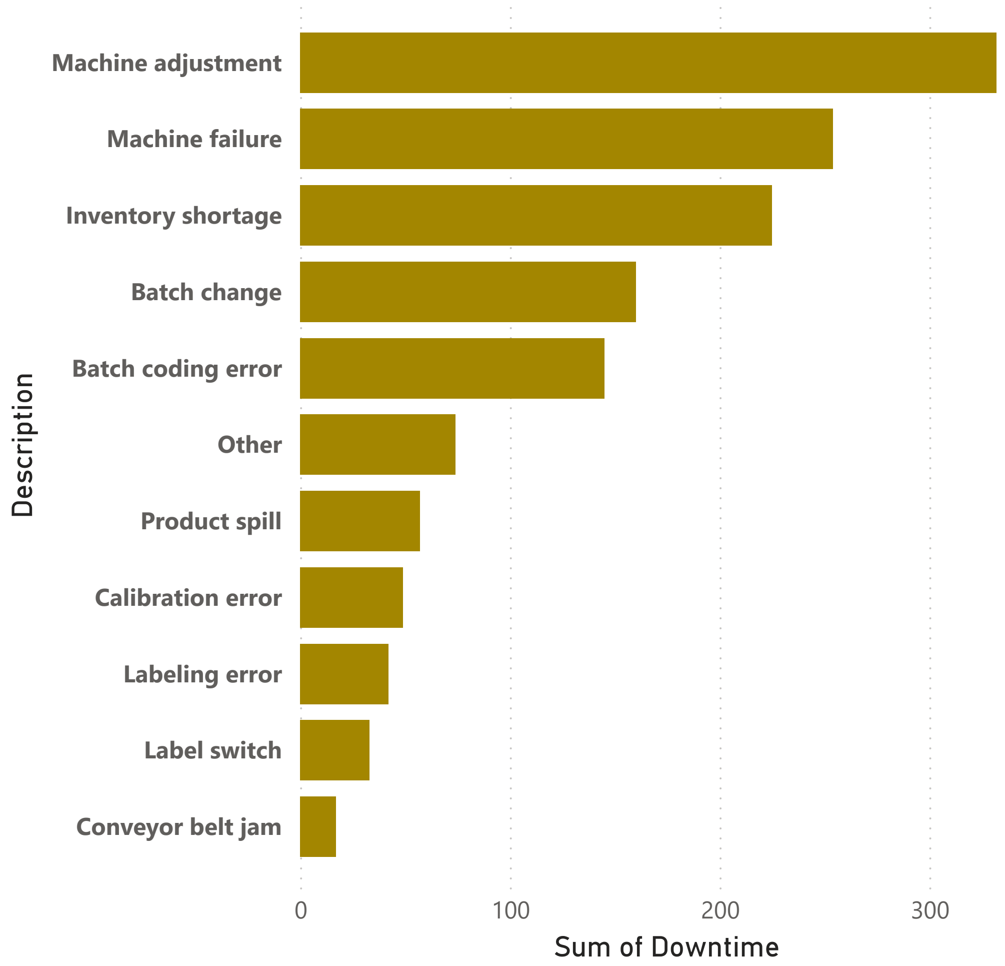
Period covered: 29 August - 3 September 2024

61	612	776	44
Downtime Frequency	Non-Operator Downtime	Operator Error Downtime	% Non Operator Error

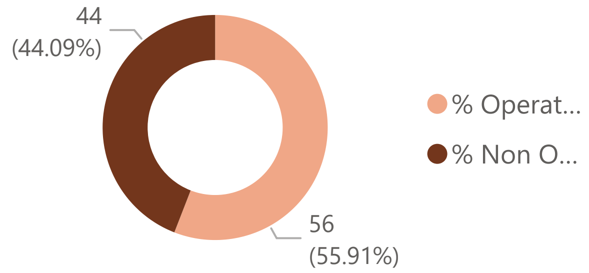
Downtime by Date



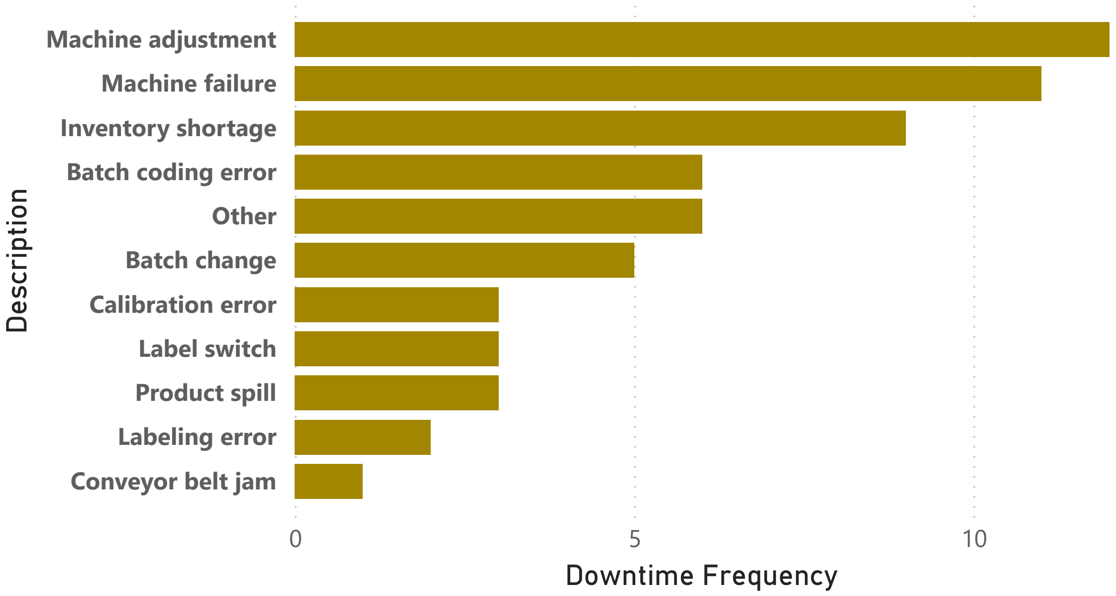
Downtime by Description



% Operator Error and % Non Operator Error



Downtime Frequency by Description



A total of **61 downtime events** were recorded, with **operator-related downtime (776 mins) exceeding non-operator downtime (612 mins)**, accounting for **56% vs 44%** respectively. Downtime fluctuated across the period, indicating inconsistent operational performance.

The main contributors to downtime were **machine adjustment (12 events, 332 mins) and machine failure (11 events, 254 mins)**, making them the most significant causes of production delays.

Overall, both human-related factors and machine issues are slowing down production, with operator errors contributing more to total downtime, while machine-related issues remain a major recurring cause.

Operator Performance Analysis:

This examines how individual operators influence productivity by assessing their contribution to efficiency, downtime, and the nature of challenges they face.

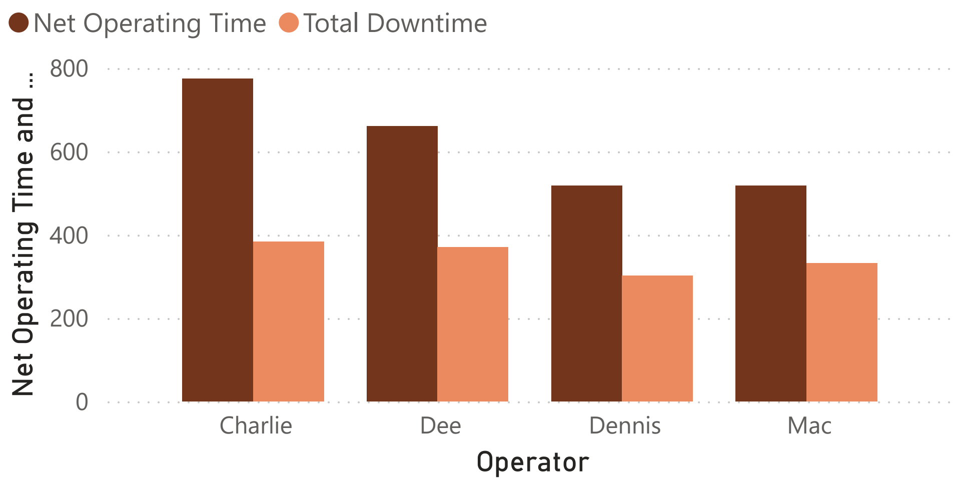
Period covered: 29 August - 3 September 2024

3858
Total Production Time

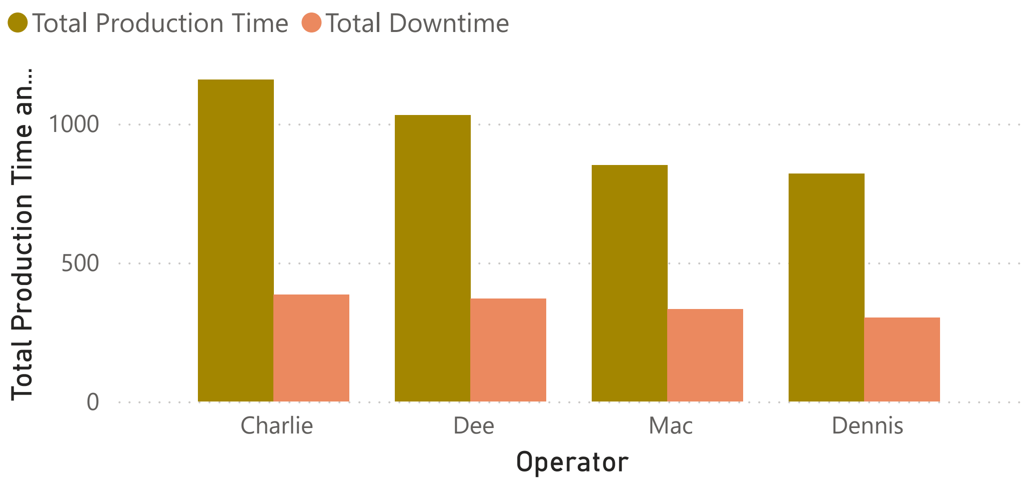
1388
Total Downtime

64%
Efficiency %

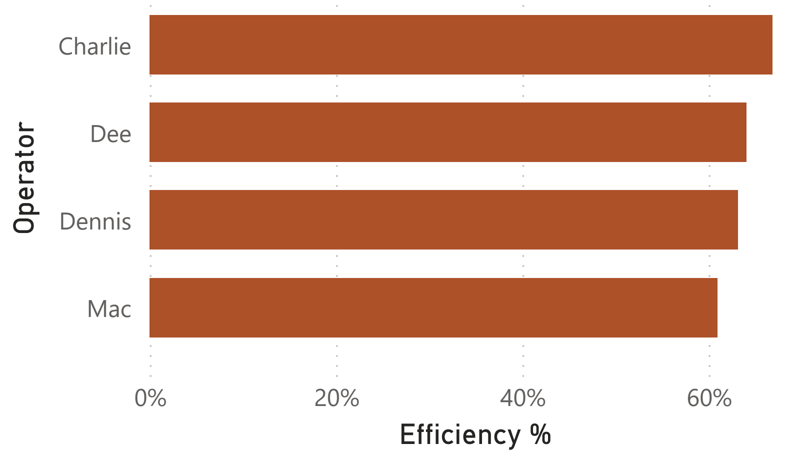
Net Operating Time and Total Downtime by Operator



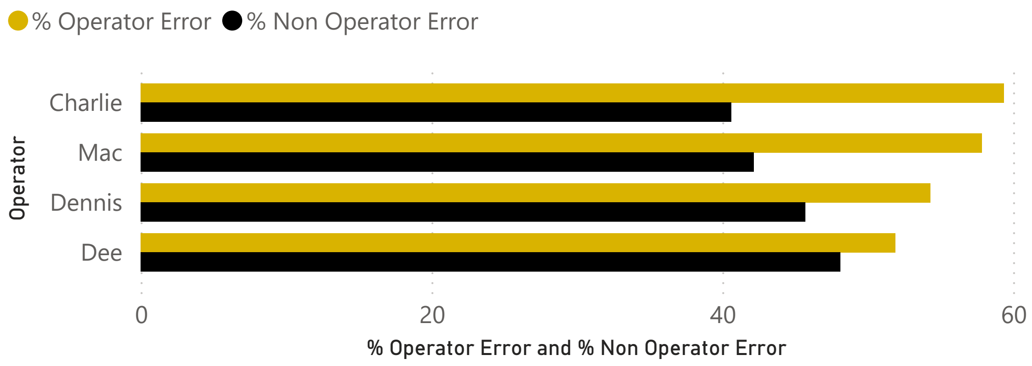
Total Production Time and Total Downtime by Operator



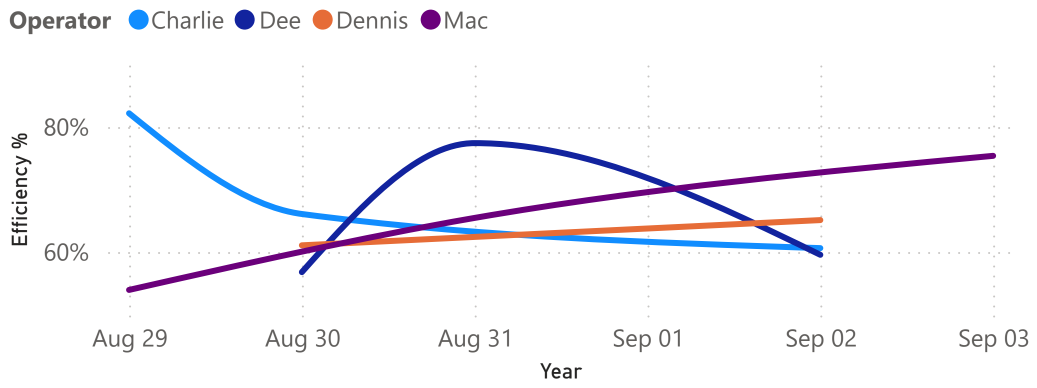
Efficiency % by Operator



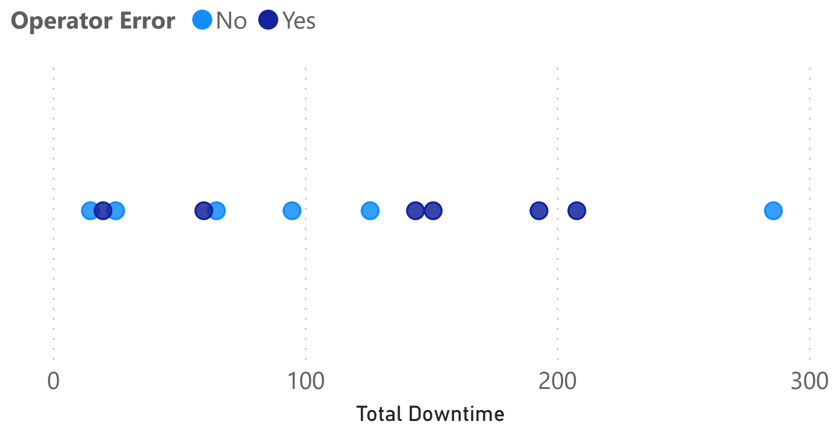
% Operator Error and % Non Operator Error by Operator



Efficiency % by Year, Quarter, Month, Day and Operator



Total Downtime by Flavor and Operator Error



Charlie handled the highest production and operating time, as well as the most downtime, followed by **Dee, Dennis, and Marc**. Despite this, Charlie achieved the **highest overall efficiency (67%)**, while **Marc recorded the lowest (61%)**. Charlie also had the highest **operator-related errors**, whereas **Dee recorded the most non-operator errors**.
At the start of the period, **Charlie led with an efficiency of 82%**, while **Marc had the lowest at 54%**. By the end, **Marc improved significantly to 75%**, while **Charlie's efficiency declined to 61%**, with other operators maintaining relatively stable performance. Overall, efficiency varies by operator, with **Charlie driving high output but also higher errors**, and **Marc showing the most improvement over time**.